

Zhifeng Gu

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Education

The Hong Kong Polytechnic University , Hong Kong	Sep 2023 – Present
• Doctor of Philosophy, Advised by Prof. Bing Wang • Research Interests: Multimodal Perception, Spatial Intelligence, World Models	
Beihang University , Beijing, China	Sep 2019 – Jun 2023
• Bachelor of Software Engineering, Advised by Prof. Lu Sheng	GPA: 3.8/4.0

Internships

XPENG Robotics , Shenzhen, China	Apr 2026 – Present
• Research Direction: Spatial Intelligence for Embodied Agents	

Publications

R³L: Reasoning 3D Layouts from Relative Spatial Relations

Zhifeng Gu*, Yuqi Wang*, Bing Wang

ICML 2026

MMOne: Representing Multiple Modalities in One Scene

Zhifeng Gu, Bing Wang

ICCV 2025

Research Experience

Spatial Reasoning for 3D Layout Generation	Jul 2025 – Jan 2026
SPACE Group, HK PolyU, Advised by Prof. Bing Wang	
• Present a general framework for 3D layout generation, featuring invariant spatial decomposition and consistent spatial imagination to improve the reliability and consistency of multi-hop relative spatial reasoning.	
Multimodal Scene Representation	Sep 2024 – Mar 2025
SPACE Group, HK PolyU, Advised by Prof. Bing Wang	
• Propose a general framework for representing multiple modalities in one scene, with a modality modeling module and a multimodal decomposition mechanism to address modality conflicts.	
6D Object Pose Estimation and Tracking	Jun 2022 – Jun 2023
VCG Lab, Beihang University, Advised by Prof. Lu Sheng	
• Study 6D object pose tracking in human-object interaction scenes, where hand occlusion poses great challenges. • Develop a motion-centric framework that explicitly models object motion across successive frames and refines pose estimation through shape completion and attention-based temporal integration.	

Selected Honors and Awards

- Outstanding Graduate of Beihang University

Selected Skills

Python, PyTorch, C/C++, CUDA, Blender, etc.